Aula Conectada

Investment case · Bolivia

Hernando Grueso, PhD
UNICEF

2026-05-01

1 · Diagnostic — gaps in education

Target population — adolescent girls 10–19 in Bolivia

Segment	Weighted N	% of girls 10–19
All adolescents 10–17 (weighted)	1,982,102	203.0%
All girls 10–17 (weighted)	976,569	100.0%
Girls 10–17 — rural	281,797	28.9%
Girls 10–17 — Indigenous	146,455	15.0%
Girls 10–17 — poor (INE)	437,692	44.8%
Girls 10–17 — disability (WG 3+)	4,142	0.4%
Girls 10–17 — no HH internet	215,766	22.1%
Girls 10–17 — priority (rural OR Indigenous OR poor OR disability)	567,836	58.1%

How to read the education indicators

Definitions (EH 2024)

The Household Survey classifies each adolescent into one of three categories regarding their educational situation:

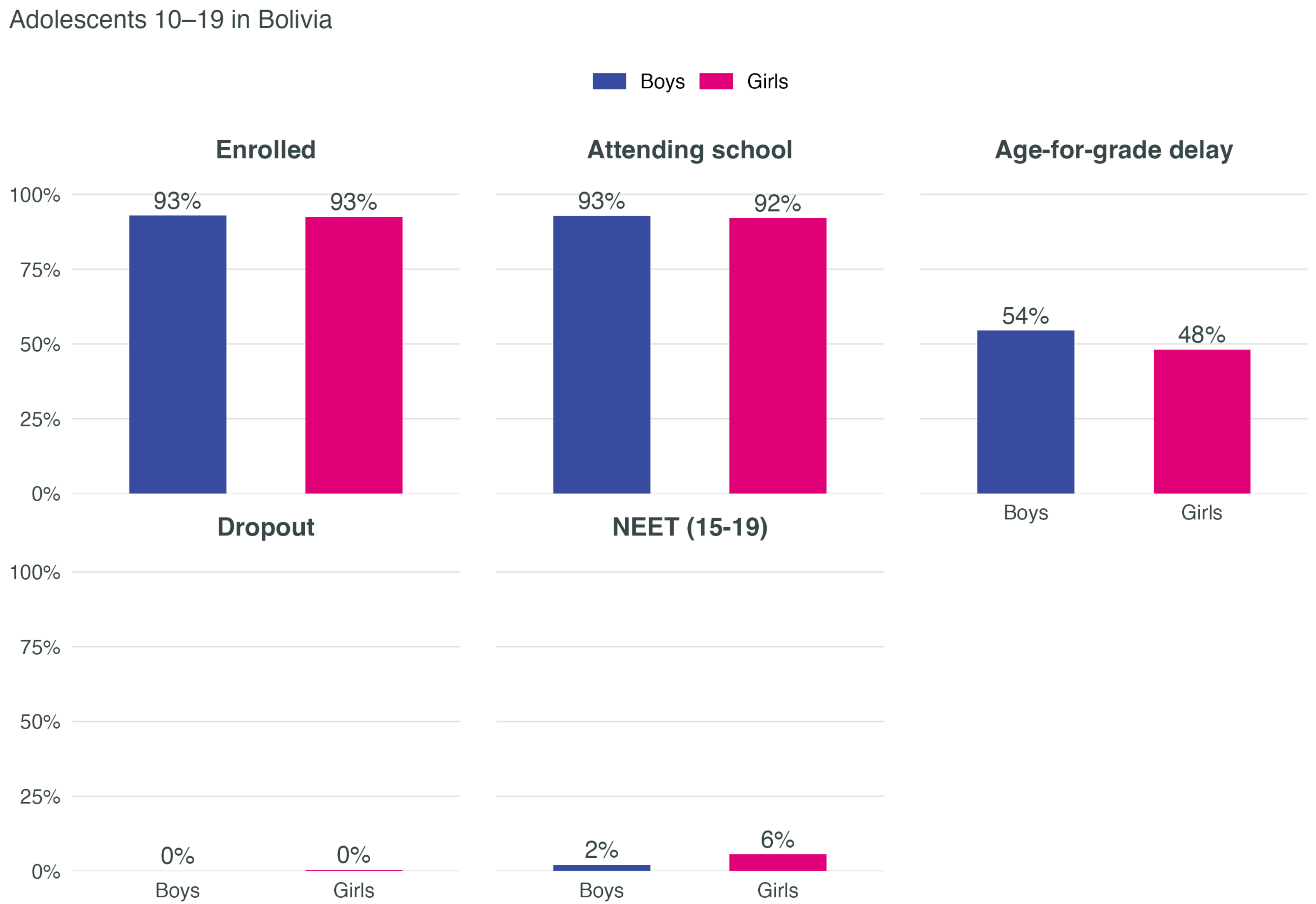
- **Enrolled** — registered for the school year, attending or not
- **Attending** — enrolled **and** currently attending classes
- **Dropout** — enrolled but no longer attending → disengagement during the year
- **Age-for-grade delay** — years of schooling below those expected for the age
- **NEET** — neither in school nor employed (15–19 years)

Why it matters

In Bolivia, formal enrolment is between 95% and 98% across nearly all subgroups. This means looking only at enrolment shows very little variation between girls/boys, urban/rural areas, or indigenous/non-indigenous groups.

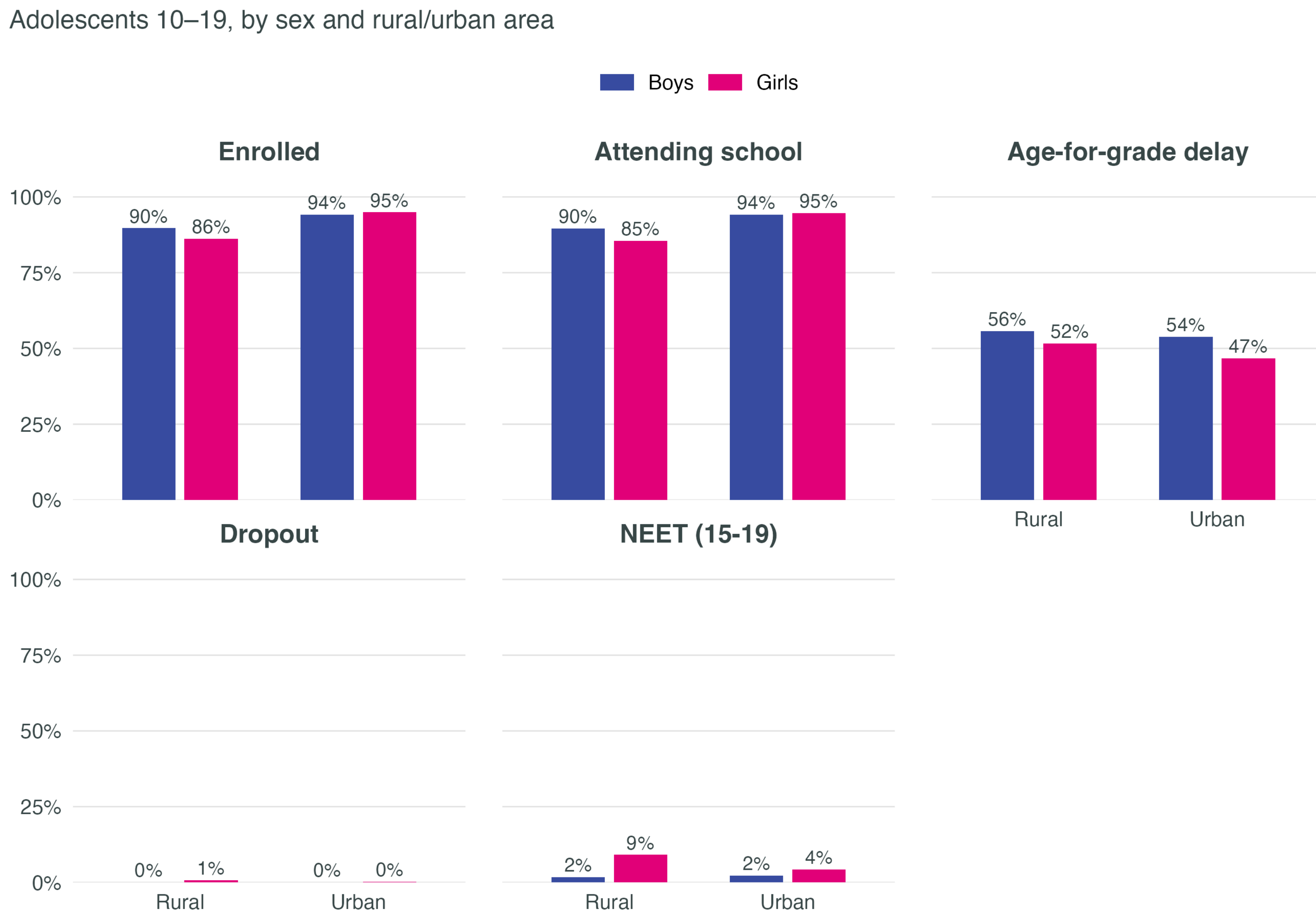
The indicators that capture where effective exclusion occurs are **attendance, dropout, age-for-grade delay, and NEET**. These reveal the territorial, indigenous, and socioeconomic gaps.

Gender gaps in education



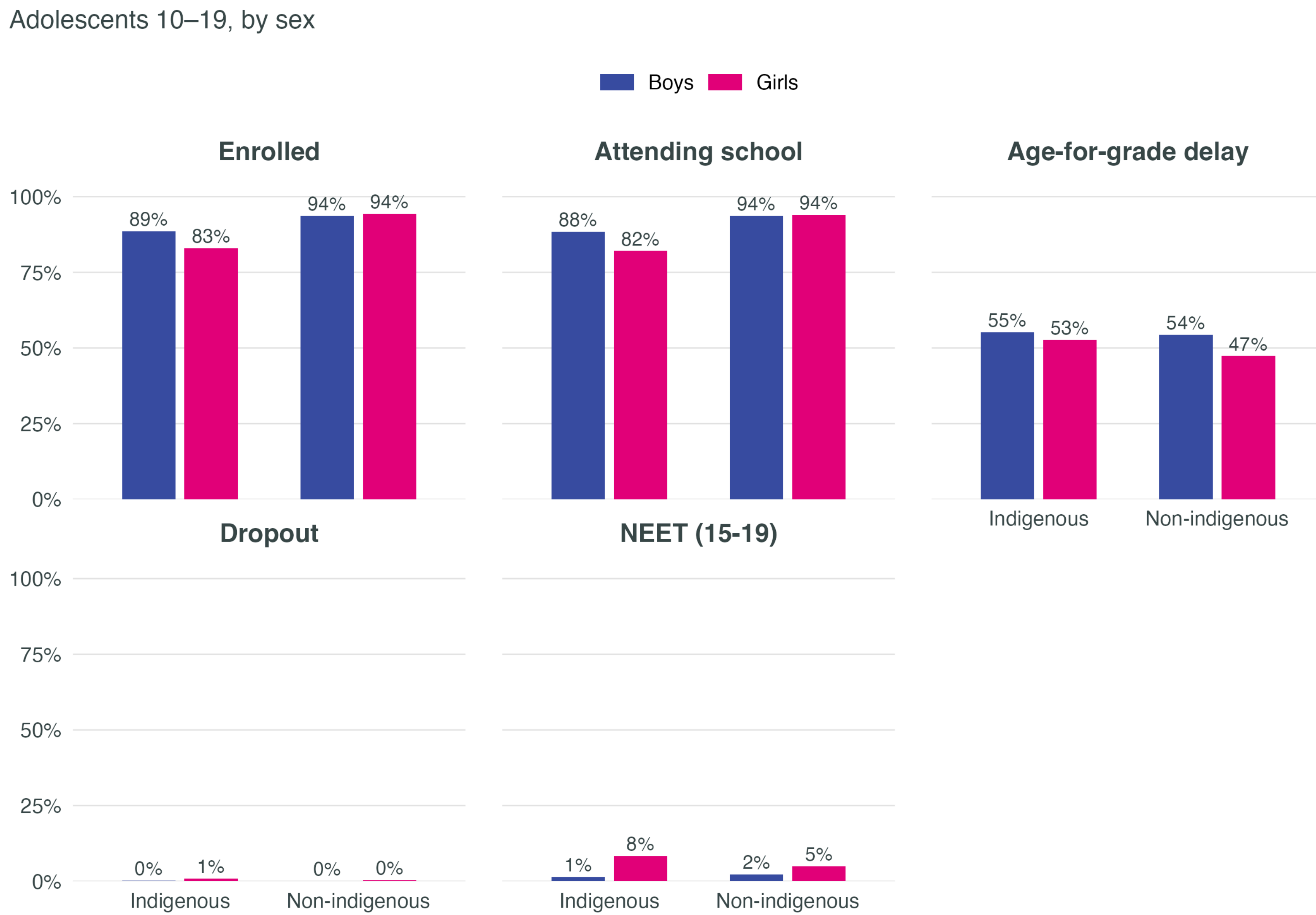
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Education outcomes by area of residence



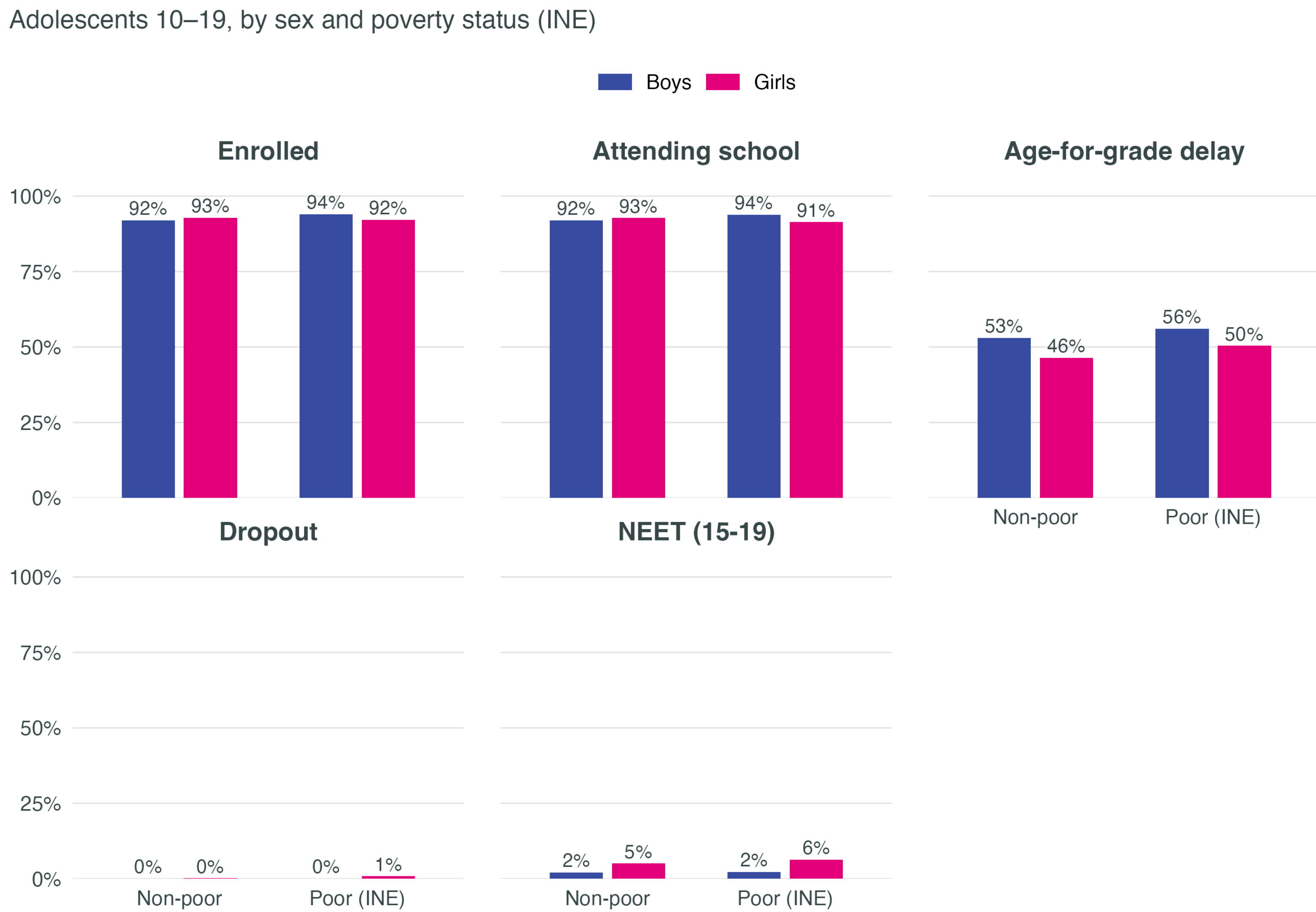
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Education outcomes by indigenous status



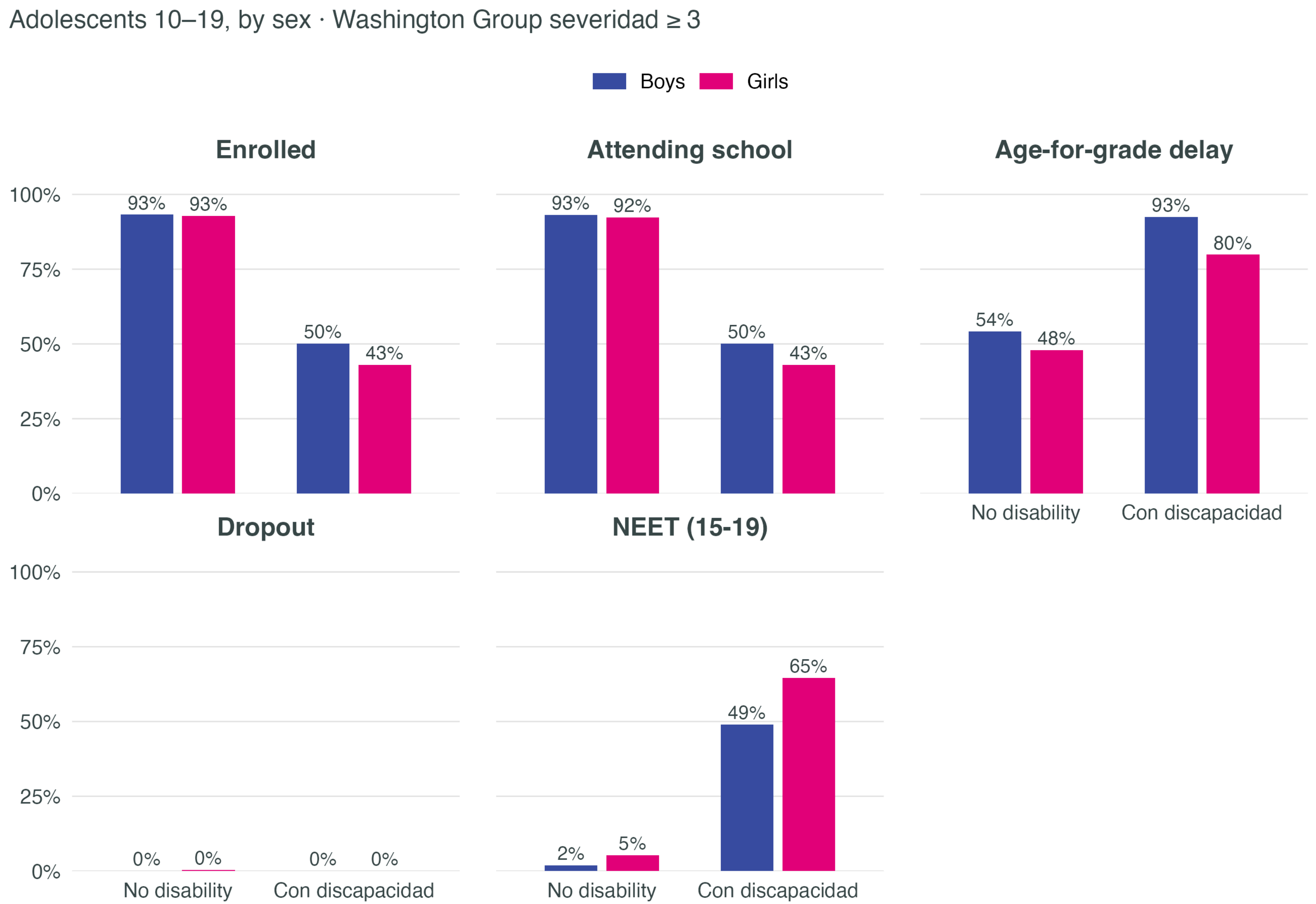
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Education outcomes by poverty status



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Education outcomes by disability



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19. Small sample size (~0.4%) — interpret with caution.

2 · Why a digital approach is needed

Digital access is a territorial and socioeconomic gap

EH 2024 data are at the household level and therefore do not allow direct measurement of girls' and boys' individual access. At the household level, differences by sex of the adolescent are minimal. **The relevant differences are observed between rural and urban households, indigenous and non-indigenous, and by poverty status.**

Dimensions of the digital divide

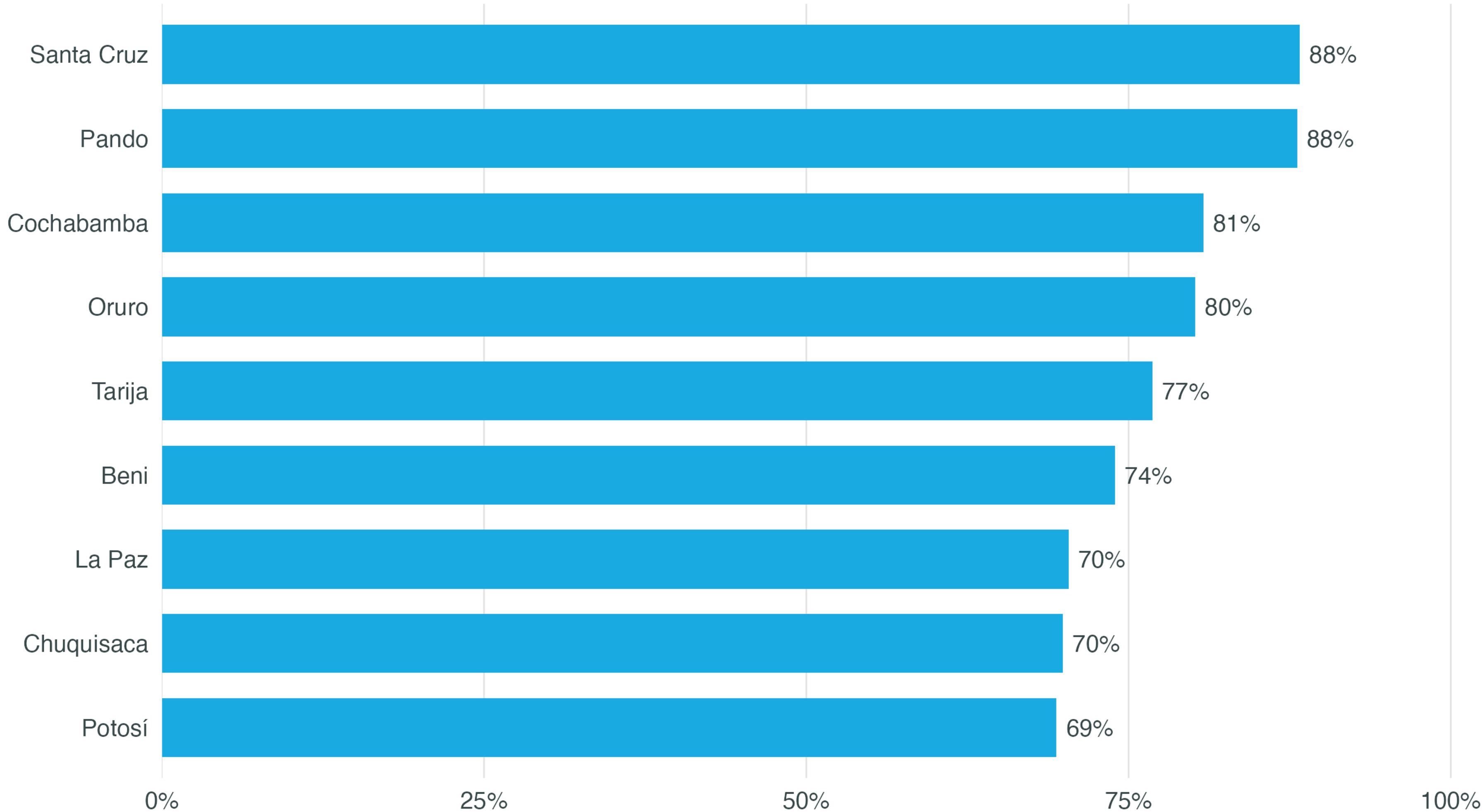
- Connectivity and infrastructure in rural areas
- Device access in indigenous households
- Economic affordability of internet and devices
- Digital skills and meaningful use

Implication for the programme

- Determining whether a gender digital divide exists at the individual level would require data on use, skills, and autonomy that EH 2024 does not capture
- At the household level, access gaps concentrate in the same subgroups that face the largest education gaps
- Aula Conectada acts on institutional access and pedagogical use, not on household access

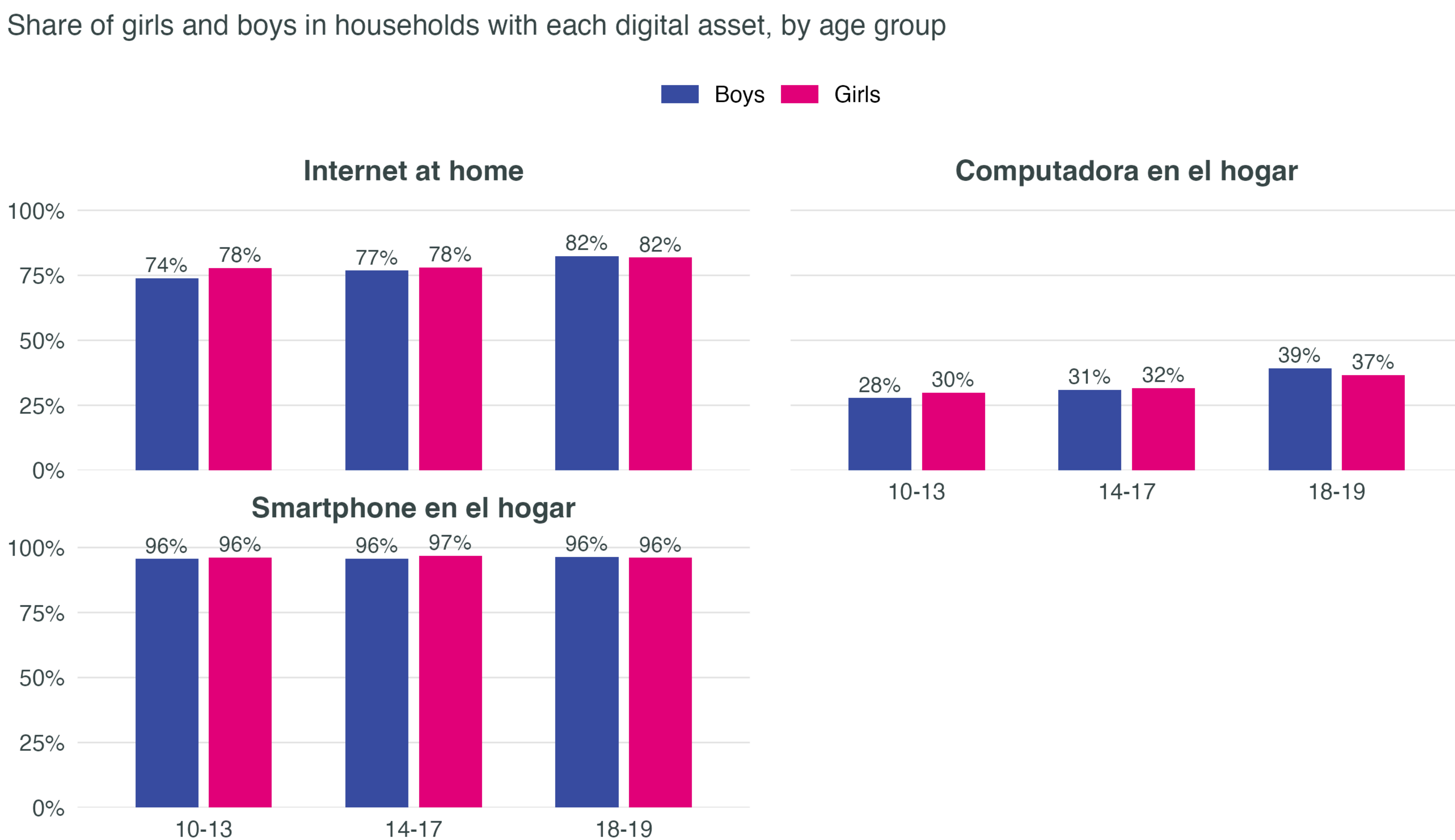
Digital divide — departmental variation

Household internet, adolescent girls 10–19, by department



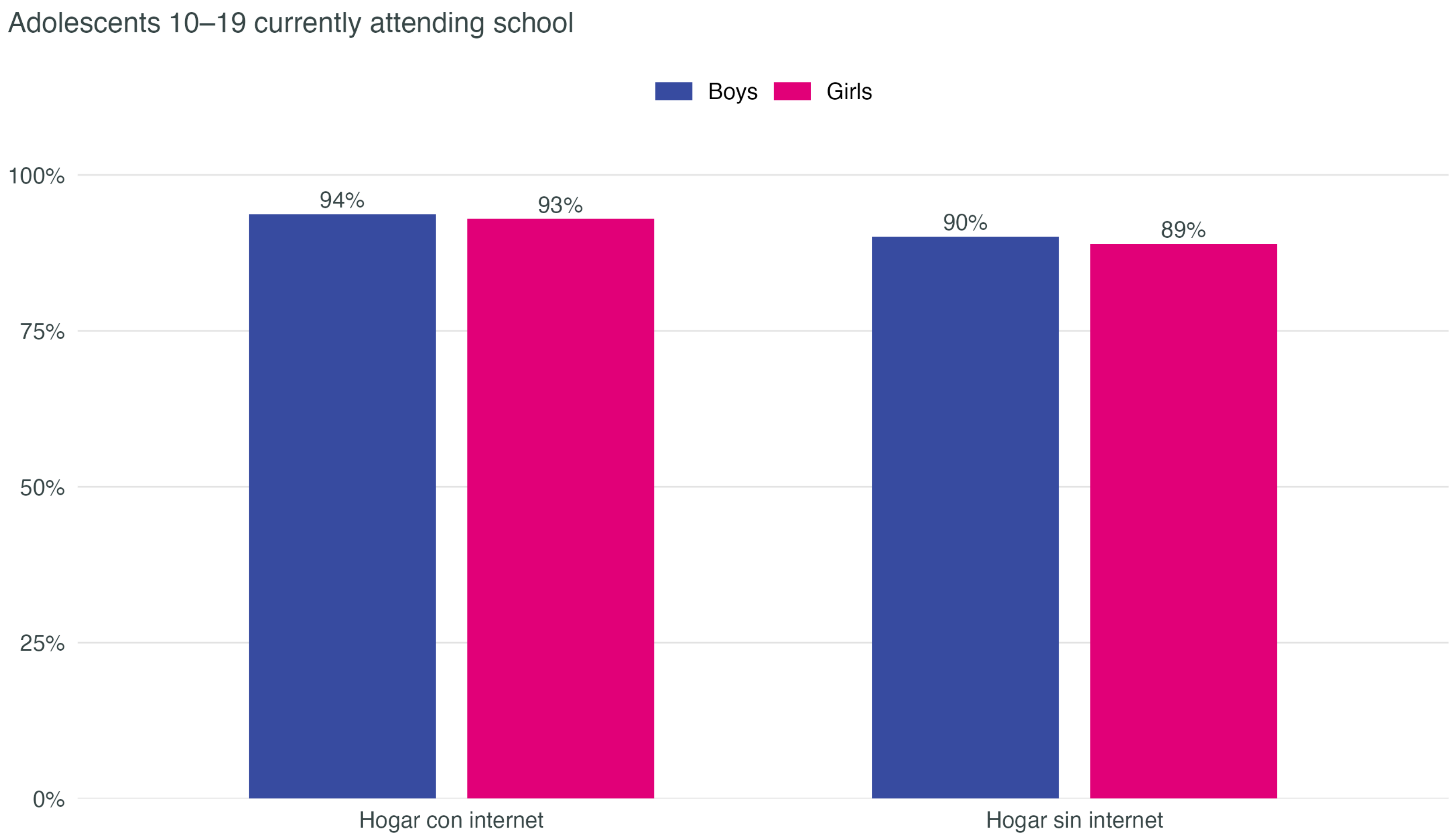
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Household digital access — no differences by sex



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Connected households show marginally higher school attendance for girls



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

3 · Multivariate analysis

Controlled gender gaps — main coefficients

Predictor	Enrolled	Attending school	Age-for-grade delay	NEET (15-19)	HH internet	Any HH device
Girl (1=yes)	-0.1 pp	-0.5 pp	-6.5 pp***	+2.1 pp***	+1.3 pp	+0.4 pp
Rural	-6.8 pp***	-7.0 pp***	+4.9 pp**	+1.4 pp*	-24.1 pp***	-4.4 pp**
Indigenous	-6.1 pp***	-6.3 pp***	+1.8 pp	+0.5 pp	-0.7 pp	-0.4 pp
Disability (WG3+)	-47.2 pp***	-46.9 pp***	+35.6 pp***	+51.1 pp***	-1.2 pp	+3.9 pp***
Poor (INE)	+1.2 pp	+0.8 pp	+5.8 pp***	+0.5 pp	-14.2 pp***	-1.6 pp*

Being a girl is not associated with a significant average gap in enrolment or attendance. It is associated with lower age-for-grade delay (−6 pp***), a pattern documented in the region and consistent with girls' greater on-time persistence through primary and early secondary education. The strong predictors across all outcomes are **rural, poverty, indigenous, and disability**. Weighted OLS with department fixed effects; standard errors clustered at the PSU level.

Gender interactions — where the gaps concentrate

Interaction × Girl	Attending school	Enrolled	dropout	NEET (15-19)	HH internet	Any HH device
Girl × Rural	-4.7 pp***	-4.3 pp**	+0.4 pp	+2.5 pp**	+2.4 pp	-0.8 pp
Girl × Indigenous	-5.6 pp**	-5.2 pp**	+0.4 pp	+2.6 pp*	+1.3 pp	-0.6 pp
Girl × Disability	-3.2 pp	-3.7 pp	-0.5 pp*	+4.3 pp	-5.0 pp	-1.4 pp
Girl × Poor	-2.6 pp	-2.0 pp	+0.6 pp	+0.9 pp	+2.5 pp	-1.3 pp

How to read: each coefficient shows how much larger (or smaller) the gender gap is in that subgroup compared to the average. For example, a rural girl faces an attendance gap that is X pp larger than the gap for an urban girl. The strongest interactions appear among **rural** and **indigenous** girls, suggesting that the programme should prioritise these subgroups for greatest impact. Significance: *p<0.10, **p<0.05, ***p<0.01.

4 · Evidence → Theory of Change

What the international evidence tells us

Consistent findings

- Isolated hardware rarely improves learning
  (De Melo et al., 2014; Cristia et al., 2017)
- Digital tools work **with structured pedagogy**
  (Gómez-Ruiz & Franco, 2023)
- Teacher training is the central mechanism
  (Olugbade et al., 2025; Bassi et al., 2016)
- Gender-responsive content improves **STEM aspirations and persistence**
  (Outes-León et al., 2020; Porter & Trzesniewski, 2020)

Parenting layer

- Parenting interventions reduce **physical and emotional violence against girls** (IRR 0.55)
  (Cluver et al., 2017 — South Africa)
- Digital parenting apps improve **non-violent practices and parent-daughter communication**
  (Janowski et al., 2025 — Tanzania)
- Restrictive gender norms at home are a structural barrier to learning, attendance and the STEM transition

Programme components — inputs and activities

Inputs

- Gender-responsive materials and safe access: digital content with female representation and free of gender bias
- STEM modules designed to retain and engage girls — a key gap in the STEM transition in Bolivia
- Connectivity and devices prioritised by need/vulnerability (rural, indigenous) rather than blanket delivery to schools

Activities

- Gender-transformative pedagogy and teacher training, with curricular integration of female role models
- Parental engagement for norm shift and linkage to STEM
- Girls' empowerment, leadership and agency — beyond mere participation
- Socio-emotional learning and life skills, based on the literature review
- Mentoring and accompaniment

Programme components — outputs and outcomes

Outputs

- Data systems tracking sex-disaggregated participation and deeper learning outcomes
- Teachers trained, classrooms equipped, and content in use
- Girls integrated in STEM activities, with priority groups explicitly identified

Short-term outcomes

- Teacher practices that promote girls' active participation
- Support for STEM transition pathways
- Strengthened digital confidence and skills
- Safe and supportive learning environments

Medium-term outcomes

- Better attendance, progression and learning
- Higher interest and persistence in STEM
- Reduced household violence against girls, through the parenting layer
- Shifts in gender norms at home and school

Long-term outcomes

- Higher transition to tertiary education, particularly in STEM
- Better employability and reduction in wage gaps

From inputs to impact — proposed Theory of Change

From inputs to impact: technology as means, pedagogy as engine

Inputs	Activities	Outputs	Short-term outcomes	Medium-term outcomes	Expected Impact
Teachers Connectivity Devices Digital content Inclusive materials STEM modules	Teacher training Digital pedagogy Collaborative learning Curriculum integration Girls' participation Family engagement	Teachers trained Classrooms equipped Content in use Girls in digital activities Priority groups identified	Teaching practice Equitable participation Digital confidence Digital skills Higher attendance	Better learning Less dropout Better progression STEM interest Less digital exclusion	Reduction of gender and intersectional gaps in digital learning and STEM trajectories

5 · Methodology — adapting evidence to the Bolivian context

How we adjust each external anchor

Each anchor study (Outes-Sanchez, Bandiera, etc.) reports a treatment effect $\hat{\beta_S}$ in its source population S. We project it to Bolivia T multiplicatively via five attenuation factors $\phi_k \in [0,1]$:

$$\hat{\beta_T} \approx \hat{\beta_S} \cdot \phi_{\text{pop}} \cdot \phi_{\text{baseline}} \cdot \phi_{\text{geo}} \cdot \phi_{\text{econ}} \cdot \phi_{\text{delivery}}$$

- ϕ_{pop} — standardised age and % rural
- ϕ_{baseline} — outcome headroom in Bolivia
- ϕ_{geo} — Haversine distance from La Paz (banded)
- ϕ_{econ} — Bolivia/anchor GDP per capita ratio
- ϕ_{delivery} — 1.0 (government) or 0.7 (NGO/research)

Uncertainty — 95% CI propagated from the anchor's reported SE. No synthetic noise added to the ϕ factors.

Structural Equation Model — synthesising the ToC

Independent of the anchor-by-anchor projections, a SEM integrates **all** anchors into a single model of the Theory of Change. Estimation pending pilot + MICS data + costing.

- **Measurement model**: latent constructs (TeacherCap, GirlSkills, HomeEnv) loading on observable indicators
- **Structural model**: ToC arrows estimated jointly, with parameters informed by the ϕ-adjusted anchors as priors
- **Output**: a single compound projection with full uncertainty propagation, useful for the cost-effectiveness analysis

Full equations in appendix A10.

6 · Anchors and projections

Direct anchors — digital education and gender norms

Programmes with measured effects **on the educational outcomes** Aula Conectada seeks to improve:

Programme	Country	External effect (95% CI)	ϕ_pop	ϕ_geo	ϕ_econ	ϕ_deliv.	n par.	Adjusted effect (95% CI)	Sig.
Mentalidad de crecimiento (Porter et al.)	Sudáfrica	-0.040 SD [-0.236, 0.156]	0.46	0.50	0.54	0.70	4	-0.003 SD [-0.020, 0.013]	—
ELA (Bandiera et al., 12-17 cohort)	Sierra Leona	7.2 pts [2.0, 12.5]	0.41	0.50	1.00	0.70	5	1.0 pts [0.3, 1.8]	✓
Plan Ceibal + programación (Gómez-Ruiz)	Uruguay	0.130 SD [0.032, 0.228]	0.49	0.85	0.30	1.00	4	0.016 SD [0.004, 0.028]	✓

Reading the CI: if zero is inside the interval, the anchor's original effect is not statistically distinguishable from zero — this reflects the source study's uncertainty, not an adjustment error. The **Sig.** column marks with ✓ the anchors that are statistically distinguishable from zero. **ELA caveat:** the high population similarity reflects overlap in observable variables (rural, age), but the post-Ebola context of Sierra Leone differs substantially.

Indirect anchors — parenting layer (via SEM)

Parenting programmes that **do not act directly** on enrolment/attendance, but on the home environment (norms, violence). They enter the SEM as parameters of the parenting → home environment → educational outcomes pathway.

Programme	Country	Original outcome	ϕ_pop	ϕ_geo	ϕ_econ	ϕ_deliv.	Model use
ParentApp Tanzania (Janowski 2025)	Tanzania	see A9 (SEM)	0.16	0.50	1.00	0.70	input parameter for parenting → home environment
Parenting for Lifelong Health (Cluver et al. 2017)	Sudáfrica	see A9 (SEM)	0.97	0.50	0.54	0.70	input parameter for parenting → home environment

Cluver et al. (2017) reports effects on physical and emotional violence (IRR 0.55, not directly translatable to enrolment). Janowski et al. (2025) reports app engagement (modules completed). Effects propagate to educational outcomes via the SEM, not directly.

Key assumptions

The five ϕ factors are claims, not estimates

- $\phi_{\text{pop}} = 1$ when the source and target are identical in age and % rural
- $\phi_{\text{baseline}} = 1$ when Bolivia has full headroom; less when the indicator is already close to its ceiling
- ϕ_{geo} is banded by Haversine distance (1.00/0.85/0.65/0.50)
- ϕ_{econ} is capped at [0.30, 1.00] for stability
- ϕ_{delivery} reflects government implementation versus NGO/research

Robustness

- Multiplicative form is **conservative**: each $\phi \leq 1$, so the projection only attenuates the external effect, never amplifies
- Anchors are **only used** if their age range overlaps ≥ 25% with the target (10–19) and at least 2 demographic variables are comparable
- Sensitivity to alternative weightings (additive, distance-decay) in technical methods paper (in preparation)

Synthesis — expected returns from Aula Conectada

Short–medium term outcomes
(priority group, ≈ 568K girls)

- **Learning** (mathematics): +0.09 SD ↔ +0.18 SD
  (Outes-León et al., 2020; Porter & Trzesniewski, 2020)
- **Computational thinking and STEM:** +0.06 SD to +0.08 SD
  (Näslund-Hadley et al., 2025)
- **Dropout averted** in higher-exclusion areas: 7–14 pp
  (Bandiera et al., 2018)

Long-term returns
(comparable regional evidence)

- **+15.2 pp** in female employment rate
  (Ñopo et al., 2008 — ProJoven Perú)
- **+93%** in female labour earnings
- **Mechanism:** early digital skills → school progression → employability

7 · Synthesis

Key findings

1. Formal enrolment in Bolivia is very high and varies little across subgroups

To identify relevant educational gaps it is necessary to look at attendance, dropout, age-for-grade delay and NEET. That is where important differences appear.

2. EH 2024 does not allow direct measurement of an individual-level digital gender gap

The data are at the household level, not the individual level. At the household level, differences by sex of the adolescent are minimal. Determining whether a gender gap exists in digital use, skills, and autonomy would require individual data that this survey does not capture.

3. The relevant digital gaps are territorial and across subgroups

At the household level, access gaps (internet, devices) concentrate between rural and urban, indigenous and non-indigenous, and across socioeconomic conditions.

4. Education gaps concentrate in the same subgroups

Girls and boys who are rural, indigenous, in poverty, or with disability face significant gaps in attendance, dropout and NEET. Multivariate analyses and gender interactions confirm these subgroups should be the programme’s priority.

Pending tasks

- **Incorporate violence indicators from MICS Bolivia 2019** to complement the parenting layer of the Theory of Change and enable SEM estimation
- **Integrate cost data** for the Aula Conectada programme when available, to build the basis for cost-effectiveness analysis
- **Estimate return on investment** once projections are complete, combining adjusted effects with programme costs

8 · Costing data

9 · Cost-effectiveness analysis

10 · Methodological Appendix

A1 · Data and sample

Source: Household Survey 2024 (EH 2024) — INE Bolivia · Collection November–December 2024 · National coverage, urban and rural · Unit of analysis: members of households in private dwellings.

Sample: 12,718 households · 39,497 persons · Analytical subsample: **6,254 adolescents 10–19** (≈ 977K girls and 1.05M boys expanded) · Files: Persona, Vivienda, Equipamiento.

Survey design: Primary sampling units (`upm`), strata (`estrato`), expansion factors (`factor`). Estimates produced with `srvyr` (PSU `lonely.psu="adjust"`).

Key variables:

- **Sex** = `s01a_02` (1=male, 2=female) → girl = (sex == 2)
- **Age** = `s01a_03` (completed years)
- **Indigenous proxy** = childhood language OR spoken language in codes 1–37, 70, 71 (Bolivian indigenous languages)
- **Poverty** = INE-derived (`poor == 1`)
- **Disability** = Washington Group, severity ≥ 3 in any domain (seeing, hearing, walking, remembering)
- **HH internet** = `s06a_19` (Housing module) · **Digital equipment** (computer, smartphone, tablet) from Equipment module

Estimation: Weighted OLS using expansion factor `factor`; standard errors clustered at PSU level (`fixest::feols`).

A2 · Empirical strategy

Base model (LPM with department fixed effects):

$$Y_{ihd} = \alpha + \beta_F F_i + \beta_R R_i + \beta_I I_i + \beta_D D_i + \beta_P P_i + \gamma X_{ihd} + \delta_d + \varepsilon_{ihd}$$

where F = Girl, R = Rural, I = Indigenous, D = Disability, P = Poor, X = controls (age).

Intersectional model with two-way interactions (× Girl):

$$Y_{ihd} = \alpha + \beta_F F_i + \gamma X_{ihd} + \sum_{k \in \{R,I,D,P\}} \theta_k \left(F_i \times k_i\right) + \delta_d + \varepsilon_{ihd}$$

Each θ_k measures how much larger or smaller the gender gap is in that subgroup. For example, θ_R captures whether the gender gap between rural girls and boys differs from the gap between urban girls and boys.

Intersectional model with three-way interactions (robustness):

$$Y_{ihd} = \alpha + \text{main effects} + \text{2-way} + \theta_{F \cdot R \cdot I} \left(F_i \times R_i \times I_i\right) + \delta_d + \varepsilon_{ihd}$$

and analogous for $F \times R \times P$ and $F \times I \times P$. Three-way interactions capture whether the gender gap in a compound intersection (e.g. rural indigenous girls) differs from what the two-way interactions would suggest separately.

Estimation by weighted OLS using `factor` (expansion factor); standard errors clustered at PSU level (`fixest::feols`).

A3 · Full coefficients table — main model

Variable	HH internet	Enrolled	Attending	Any HH device	Age-for-grade delay
Girl (1=yes)	0.013	-0.001	-0.005	0.004	-0.065***
	(0.011)	(0.008)	(0.008)	(0.005)	(0.014)
Rural	-0.241***	-0.068***	-0.070***	-0.044**	0.049**
	(0.039)	(0.020)	(0.020)	(0.022)	(0.021)
Poor (INE)	-0.142***	0.012	0.008	-0.016*	0.058***
	(0.022)	(0.008)	(0.009)	(0.008)	(0.016)
Indigenous	-0.007	-0.061***	-0.063***	-0.004	0.018
	(0.036)	(0.019)	(0.019)	(0.016)	(0.024)
Disability (WG3+)	-0.012	-0.472***	-0.469***	0.039***	0.356***
	(0.069)	(0.066)	(0.066)	(0.011)	(0.052)
N	7,517	7,517	7,517	7,517	7,517
R^2	0.150	0.141	0.140	0.034	0.058
Dept FE	Yes	Yes	Yes	Yes	Yes
Weights	Yes	Yes	Yes	Yes	Yes
Cluster SE (PSU)	Yes	Yes	Yes	Yes	Yes

Coefficients in decimal points (× 100 for pp). Robust SE in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

A4 · Full intersectional table

Interaction	Attending	Enrolled	HH internet	Any HH device
Girl × Rural	-0.047***	-0.043**	0.024	-0.008
	(0.018)	(0.018)	(0.030)	(0.013)
Girl × Indigenous	-0.056**	-0.052**	0.013	-0.006
	(0.022)	(0.023)	(0.037)	(0.013)
Girl × Disability	-0.032	-0.037	-0.050	-0.014
	(0.137)	(0.137)	(0.132)	(0.019)
Girl × Poor	-0.026	-0.020	0.025	-0.013
	(0.016)	(0.016)	(0.028)	(0.012)
N (each model)	7,517	7,517	7,517	7,517

Each row is the coefficient on the (Girl × subgroup) interaction in a separate regression. Coefficients in decimal points. Standard errors in parentheses. Robust SE clustered at PSU.

A5 · Heterogeneity analysis — regression forest

We complement the LPM with a regression forest (`grf::regression_forest`, Athey & Wager 2019) that estimates predicted outcomes conditional on the full set of demographic variables.

- **Training sample:** all adolescents 10–19 (not only girls) — lets the forest learn the gender effect endogenously
- **Forest:** `grf::regression_forest` with `honesty = TRUE` and 5,000 trees
- **Output:** variable importance and predicted subgroup gaps

The forest is **diagnostic, not causal** — there is no treatment in EH 2024 to estimate CATEs. Its value: identifying the demographic profiles where the (predicted) gap is largest, to guide programme targeting.

A6 · Bolivia-adjusted projection methodology

Algorithm for each anchor:

1. Compute ϕ_{pop} as $1 - \sqrt{d^2_{\text{age}}, d^2_{\text{rural}}}$, where d are standardised differences between source and Bolivia
2. Compute $\phi_{\text{baseline}} = (Y_{\max} - Y_{\text{Bol}})/(Y_{\max} - Y_{\text{anchor}})$ when measurable (otherwise 1)
3. Compute ϕ_{geo} in Haversine bands from La Paz: ≤1,000 km → 1.00; 1,000–5,000 km → 0.85; 5,000–10,000 km → 0.65; >10,000 km → 0.50
4. Compute ϕ_{econ} as the Bolivia/anchor GDP per capita ratio, capped at [0.30, 1.00]
5. Compute ϕ_{delivery}: 1.0 if government implementer; 0.7 if NGO/research (Vivalt 2020)
6. Multiply: $\hat{\beta_T} = \hat{\beta_S} \cdot \prod \phi_k$
7. Propagate SE: $\text{SE}(\hat{\beta_T}) = \text{SE}(\hat{\beta_S}) \cdot \prod \phi_k$
8. 95% CI: $\hat{\beta_T} \pm 1.96 \cdot \text{SE}(\hat{\beta_T})$

Exclusion rules — age overlap <25% with 10–19, or <2 demographic variables comparable → anchor not projected.

Uncertainty — 95% CI propagated from the anchor SE. No synthetic noise on ϕ factors.

A7 · Formal foundation of the transportability approach

We assume that the mismatch between the source population S (the anchor) and the target T (Bolivia) decomposes into K observable dimensions, with multiplicative attenuation:

$$\hat{\beta_T} \approx \hat{\beta_S} \cdot \prod_{k=1}^{K} \phi_k(S, T), \qquad \phi_k \in [0, 1]$$

Glossary of ϕ_k factors:

- ϕ_{pop} — standardised age and % rural
- ϕ_{baseline} — outcome headroom
- ϕ_{geo} — Haversine distance from La Paz
- ϕ_{econ} — GDP per capita ratio
- ϕ_{delivery} — 1.0 (government) or 0.7 (NGO/research)

Assumptions:

- (A1) **Sufficient observables**: the 5 dimensions capture the relevant effect modifiers
- (A2) **Separability**: effects are approximately multiplicative
- (A3) **Monotonicity**: mismatch only attenuates, never amplifies
- (A4) **SE independence**: anchor variance propagates without perturbing the ϕ_k

References: Pearl & Bareinboim (2014, Statistical Science); Stuart, Cole, Bradshaw & Leaf (2011, J. Royal Stat. Soc.); Dahabreh et al. (2020, Biometrics); Vivalt (2020, J. Eur. Econ. Assoc.).

A8 · Consolidated evidence — anchors in use

Programme	Country	Age	Effect	Implementer	ToC arrow
Mentalidad de crecimiento (Outes & Sanchez, full)	Perú	12–14	0.054 (0.030) SD	Government	Digital education
Mentalidad de crecimiento (Outes & Sanchez, reg…	Perú	12–14	0.135 (0.051) SD	Government	Digital education
Mentalidad de crecimiento (Porter et al.)	Sudáfrica	13–16	-0.040 (0.100) SD	NGO/Research	Digital education
ELA (Bandiera et al., 12-17 cohort)	Sierra Leona	12–17	7.250 (2.700) score_pts	NGO/Research	Gender norms
Irûmi — pensamiento computacional (Näslund-Hadley)	Paraguay	7–8	0.089 (0.052) SD	Government	Digital education
Plan Ceibal SOLO HARDWARE (De Melo)	Uruguay	8–12	0.000 (0.050) SD	Government	Digital education
Plan Ceibal + programación (Gómez-Ruiz)	Uruguay	14–17	0.130 (0.050) SD	Government	Digital education
ProJoven (Ñopo et al.)	Perú	16–25	0.152 (0.060) ppts	Government	Labour market
ParentApp Tanzania (Janowski 2025)	Tanzania	0–17	0.130 (0.060) SD	NGO/Research	Parenting
Parenting for Lifelong Health (Cluver et al. 2017)	Sudáfrica	10–17	0.100 (0.050) ppts	NGO/Research	Parenting

Complete list of external anchors used in the Bolivia projection (see `05_projections.R` and `R/anchors_demographics.R`). The full systematic review of 52 papers is in `data/lit_review_papers.csv`. The **ToC arrow** column indicates which arc of the Theory of Change each anchor feeds.

A9 · Robustness — three-way interactions (OLS)

To verify the heterogeneity detected by the regression forest, we estimate three-way interactions with OLS. If the forest correctly detects compound subgroups, the three-way coefficients should be significant for the outcomes with greatest heterogeneity.

Three-way interaction	Attending school	Enrolled	HH internet
Girl × Rural × Indigenous	-0.061 (0.045)	-0.052 (0.046)	0.050 (0.073)
Girl × Rural × Poor	-0.003 (0.036)	-0.008 (0.038)	-0.096 (0.069)
Girl × Indigenous × Poor	-0.002 (0.050)	0.005 (0.051)	0.096 (0.077)

How to read: each cell shows the three-way interaction coefficient and its standard error in parentheses. Significance: * p<0.10, ** p<0.05, *** p<0.01. Where no asterisks appear, the coefficient is not statistically distinguishable from zero at conventional levels. **Interpretation:** in this sample, the three-way interactions are not significant, indicating that the compound effect of belonging to multiple vulnerable groups (e.g. rural indigenous girl) is not substantially larger than the sum of effects of each group separately. Gaps accumulate roughly additively. This suggests well-targeted interventions on any axis (rural, indigenous, poverty) will reach girls who fit those characteristics.

A10 · Structural Equation Model (SEM) — proposed

Next step to integrate all anchors into a single model of the Theory of Change. We pre-register the structure here; estimation will use pilot data, MICS Bolivia 2019 and costing.

Measurement model

$$\text{TeacherCap} = \lambda_1 \text{TrainHrs} + \lambda_2 \text{DigCompEdu}$$

$$\text{GirlSkills} = \lambda_3 \text{CompThink} + \lambda_4 \text{DigConfidence}$$

$$\text{HomeEnv} = \lambda_5 \text{ParentEng} + \lambda_6 \text{GenderNorm}$$

Structural model

$$\text{ClassPractice} = \beta_1 \text{TeacherCap}$$

$$\text{GirlSkills} = \beta_2 \text{ClassPractice} + \beta_3 \text{HomeEnv}$$

$$\text{Attendance} = \beta_4 \text{GirlSkills} + \beta_5 \text{HomeEnv}$$

$$\text{Progression} = \beta_6 \text{Attendance} + \beta_7 \text{GirlSkills}$$

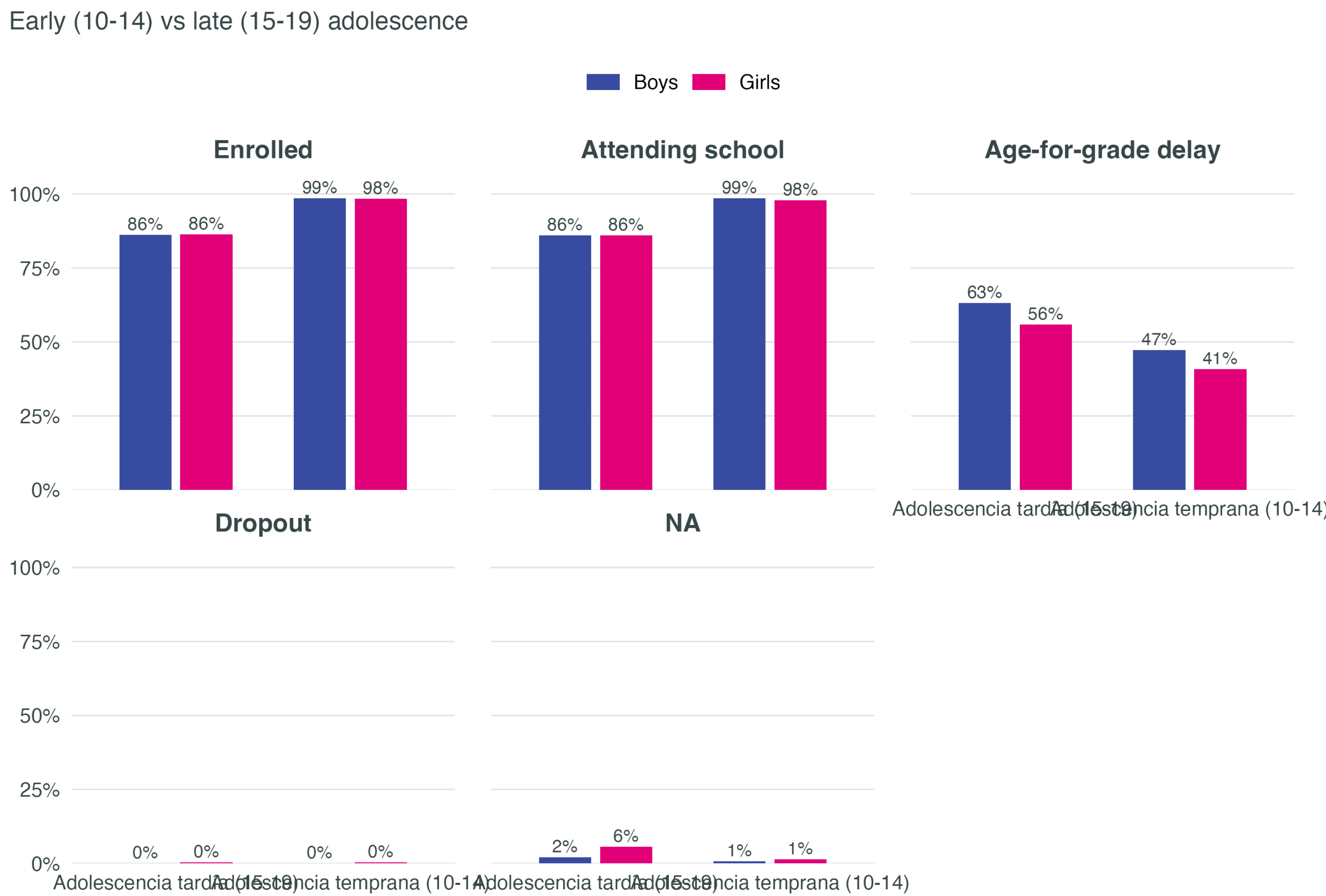
Calibration from ϕ-adjusted anchors

- $\beta_2 \leftarrow$ Outes-Sanchez + Gómez-Ruiz
- $\beta_3 \leftarrow$ ParentApp (Janowski) + Cluver
- $\beta_4 \leftarrow$ ELA (Bandiera)
- $\beta_6 \leftarrow$ measurable with EH 2024 + pilot

Identification assumptions

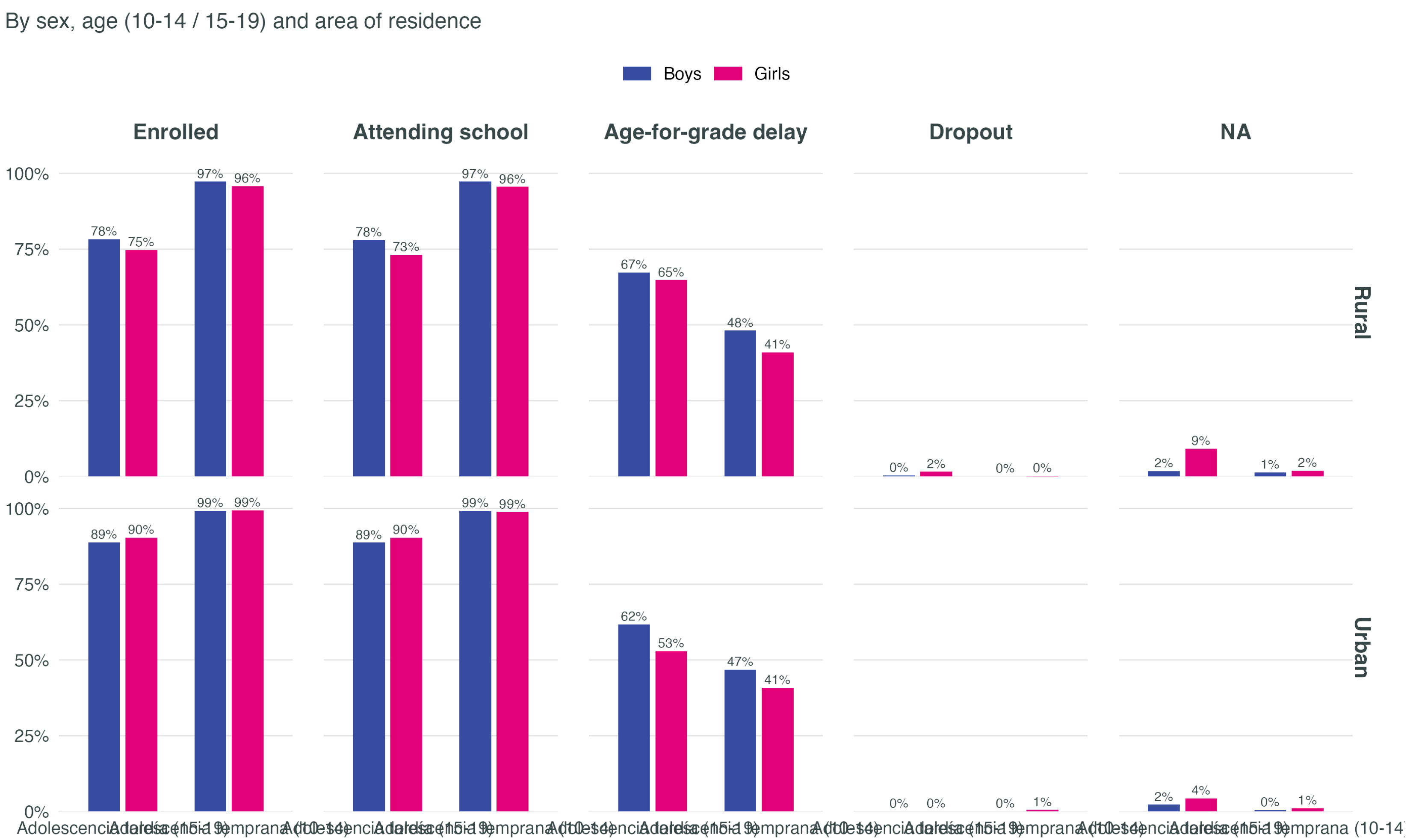
1. Independent latent errors
2. Latent scales fixed via $\lambda_1 = 1$ per construct
3. Structural parameters taken as informative (Bayesian priors with `blavaan`) to propagate uncertainty

A11 · Education outcomes by age band — overall



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19. NEET = not in school nor employed. By construction, N

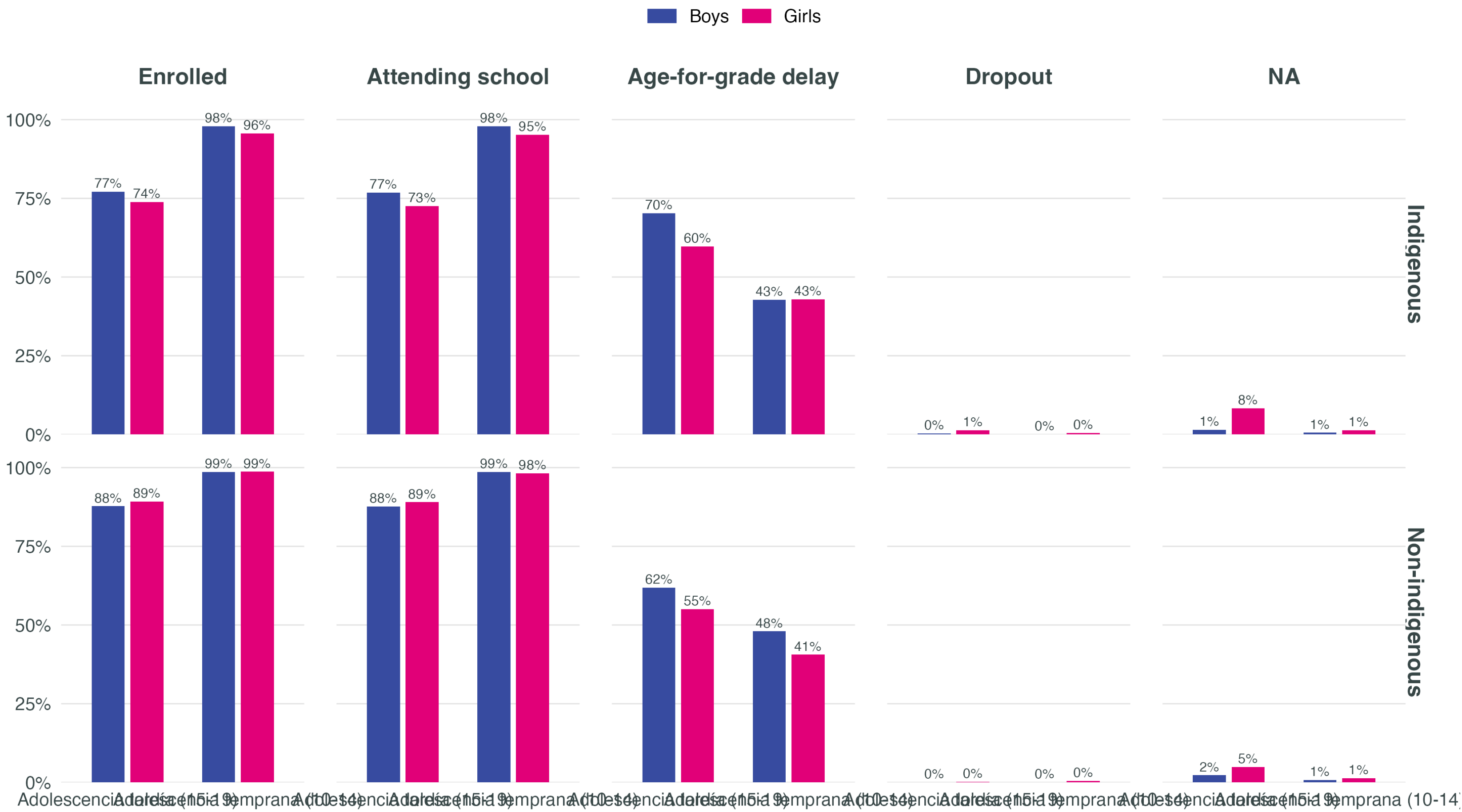
A12 · Education outcomes by age band — area of residence



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19. NEET = not in school nor employed. By construction, NEET is very low in 10-14 since n

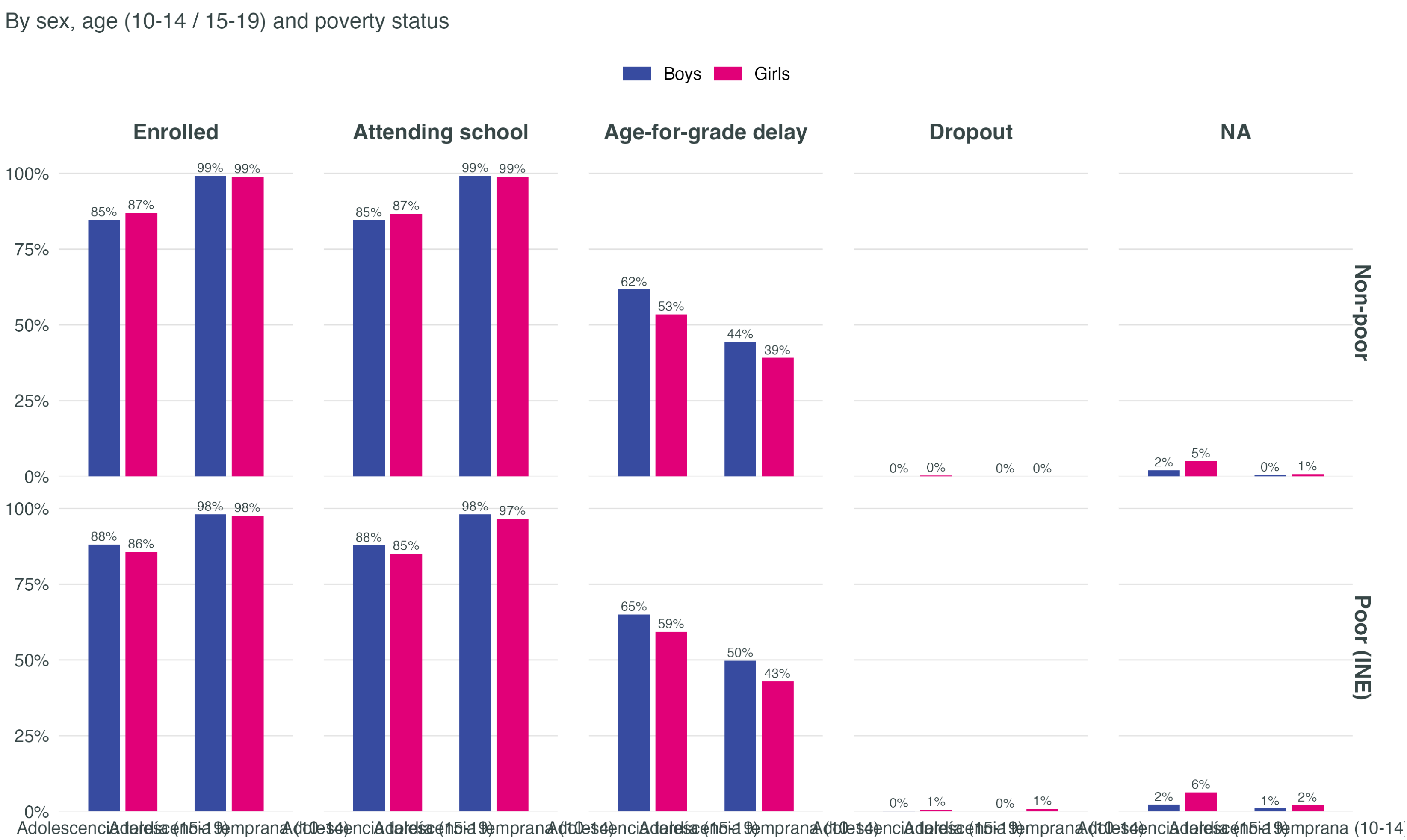
A13 · Education outcomes by age band — indigenous status

By sex, age (10-14 / 15-19) and indigenous status



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19. NEET = not in school nor employed. By construction, NEET is very low in 10-14 since n

A14 · Education outcomes by age band — poverty status



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19. NEET = not in school nor employed. By construction, NEET is very low in 10-14 since n

A15 · Education outcomes by age band — disability

A16 · Limitations

- EH measures **access to devices and internet**, not actual digital skills
- Household access **does not equal meaningful use** by the adolescent
- Disability sample size is **small** — interpret with caution
- **Cross-sectional** data: temporal trajectories cannot be observed
- Need for **direct measurement at the pilot level** to validate and refine projections

hgrueso@unicef.org